Resolution Enhancement for Forwarding Looking Multi-Channel SAR Imagery With Exploiting Space–Time Sparsity

Jingyue Lu, Lei Zhang, *Member, IEEE*, Shaopeng Wei, and Yachao Li, *Member, IEEE*

***Abstract*— Forward-looking multi-channel synthetic aperture radar (FLMC-SAR) is of the capability to achieve unambiguous 2-D images in the forward-looking slight direction. FLMC-SAR imagery usually suffers from relatively low spatial resolution as only limited Doppler diversity can be generated from the synthetic aperture. In this article, a sparsity-driven resolution enhancement algorithm is proposed to improve the resolution FLMC-SAR image of the forward-looking area. Different from conventional beamforming processing to resolve the FLMC-SAR left–right ambiguity, a Bayesian sparsity reconstruction optimization is developed for jointly ambiguity resolving and resolution enhancement in the azimuth angle image domain. The spatial structure of the target in the preliminary image domain is used as the signal sparsity with prior information to solve the constrained optimization problem for FLMC-SAR image resolution enhancement. A local least square estimator of the prior noise and signal statistics in the FLMC-SAR nonisotropic image is established in terms of determining the sparsity weight parameter. Extensive simulation and real FLMC-SAR data experiments confirm that the proposed algorithm is capable of achieving the unambiguous and resolution-enhanced FLMC-SAR image.**

***Index Terms*— Bayesian, Doppler ambiguity, forward-looking multi-channel synthetic aperture radar (FLMC-SAR), image resolution enhancement, sparsity based, sparsity driven.**

I. Introduction

DUE to the all-weather, day and night, long-distance, high-resolution, and 2-D imaging capabilities, synthetic aperture radar (SAR) [1], [2], [3], [4], [5], [6] has been widely used in military and civilian fields. The geometric configuration of SAR is flexible. Therefore, there are various working modes of SAR, such as side-looking SAR and squint-looking SAR. The imaging area is usually located on the side-looking area or squinted-looking area of the moving platform. When the antenna beam points to the area in front of the trajectory, SAR works in the forward-looking mode. In this mode, the Doppler gradient of the forward-looking slight direction decreases dramatically, resulting in low azimuth resolution in the forward-looking area. Therefore, the forward-looking area is usually regarded as the close-blind zone in the conventional SAR system. In addition, in the forward-looking SAR, the targets that are symmetrical with respect to the trajectory have identical Doppler history, inducing the left–right Doppler ambiguity [7], [8] in the forward-looking SAR imagery. In some practical applications, such as the blind landing of aircraft in severe weather and battlefield reconnaissance, it is urgent and significant to generate forward-looking SAR images with high resolution. Improving the azimuth resolution of the forward-looking area is a challenging and crucial task in SAR imagery.

Sector imaging radar for enhanced vision (SIREV) [9], [10] is the earliest forward-looking SAR imaging system, which uses an array of real apertures in the cross-trajectory direction to achieve forward-looking resolution enhancement. However, the forward-looking resolution of the SIREV system is excessively dependent on the real aperture, which results in a higher system complexity for high-resolution requirements. The geometric configuration of bistatic SAR [11], [12], [13], [14], [15] is more flexible than monostatic SAR to realize forward-looking imaging. Bistatic SAR needs to be additionally equipped with a receiver or transmitter platform to provide forward-looking Doppler diversity in order to achieve high cross-range resolution. Monopulse radar [16] imaging can realize azimuth focus and improve the visibility of the forward-looking area with low system complexity. However, due to the limitation of the monopulse angle measurement principle, it is impossible to distinguish multiple targets in the same resolution unit, which decreases the imaging quality of complicated scenes. With appropriate geometry configurations, multi-channel SAR [17], [18], [19], [20], [21], [22] is capable of reconstructing the image of the forward-looking area, which we call forward-looking multi-channel SAR (FLMC-SAR) [7], [8]. However, only limited Doppler diversity is synthesized by the long trajectory in the forward-looking area, which makes it difficult to achieve high resolution of FLMC-SAR in principle.

Sparsity-based algorithms [23], [24], [25], [26], [27], [28], [29], [30] have attracted great interest and show great potential in the field of forward-looking SAR image reconstruction. The prior information of sparsity is used to recover the high-resolution image from limited measurements. The basic

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framework of the sparse reconstruction algorithm is to exploit the sparsity of signal to improve image quality via ℓ_1 optimization [31], [32], [33]. In [2], a feature-enhanced approach based on a regularized reconstruction of the scattering field is proposed, which combines a tomographic model of the SAR observation process with the prior information regarding the nature of the features of interest. However, those sparsity-based algorithms do not consider Doppler ambiguity and limited Doppler diversity in forward-looking imaging, so it is difficult to straightforwardly apply current resolution enhancement algorithms [34], [35], [36], [37], [38] in the FLMC-SAR system.

In this article, sparsity-based resolution-enhanced imagery for FLMC-SAR is proposed to enhance the azimuth resolution of the forward-looking area. We first obtained a relatively low spatial resolution image of the forward-looking area via the SAR imaging algorithm. Due to the Doppler ambiguity in the FLMC-SAR system, the imaging result is symmetrical about the scene center. Different from conventional beamforming processing to resolve the FLMC-SAR left–right ambiguity, a Bayesian sparsity reconstruction optimization is developed to jointly achieve ambiguity resolving and resolution enhancement in the azimuth angle image domain. By analyzing the Doppler ambiguity in the FLMC-SAR imaging process, the sparse signal model of the target is established in the FLMC-SAR image domain. The spatial structure of the target in the preliminary image domain is used as the prior information of the signal sparsity to solve the constrained optimization problem for FLMC-SAR image resolution enhancement. Therefore, we convert the FLMC-SAR image resolution enhancement into sparsity-driven optimization in the Bayesian statistics sense. A local least square estimator of the prior noise and signal statistics is established to determine the sparsity weight parameter. Extensive simulation and real FLMC-SAR data experiments confirm that the proposed algorithm is capable of achieving the unambiguous and resolution-enhanced FLMC-SAR image. The main contribution of the proposed algorithm can be concluded as follows.

1) To perform azimuth resolution enhancement of forward-looking SAR scene image, the space–time cascade processing framework is developed. The separated signal has strong spatial sparsity in the space–time observation model. Furthermore, different spatial steering vectors reflected in space–time observation model, which paves an effective way for joint ambiguity resolving and resolution enhancement by exploiting space–time sparsity effectively.
2) Due to the special imaging geometry, forward-looking SAR imaging results are nonisotropic, which brings difficulties for the determination of optimal values for parameters/hyperparameters in a sparsity-driven imaging method. An adaptive local prior noise and signal statistics estimator is developed to overcome the parameters selection problem, which applies the spatial structure information of targets in the isotropic image to the sparsity-driven optimization. Resolution enhancement performance is effectively improved due to the optimization based on the targets' distribution.

This article is organized as follows. Section II gives the geometry model for the FLMC-SAR. By analyzing joint FLMC-SAR imaging and Doppler ambiguity, we present the sparse model of the FLMC-SAR in Section III. Section IV formulates a constrained optimization problem, and the proposed FLMC-SAR imaging method for enhancing the resolution of the forward-looking area is described in detail. The experimental results based on both simulated and real data are given in Section V. Section VI concludes this work.

II. FLMC-SAR Sparse Model

A. FLMC-SAR Signal Model

The geometry model for spotlight-mode FLMC-SAR is shown in Fig. 1. The radar system uses a single antenna as the transmitting antenna and a uniform linear array antenna as the receiving antenna. Setting the projection of the aperture center on the ground as the origin O, the direction of the nominal trajectory as the X-axis, the array distribution direction as the Y-axis, and the vertical direction up to the ground as the Z-axis, a Cartesian coordinate system $OXYZ$ is established. The receiver array antenna is symmetrically distributed along the Y-axis over Y_1Y_2, and the ith array element is represented by the symbol R_{Xi}. The length of the antenna's real aperture is D, and the antenna element spacing is d. The transmitter antenna is located at the midpoint of the receiver array antenna Y_1Y_2, represented by the symbol R_T. The radar is mounted on a moving platform at a height of h above the ground and moves along the nominal trajectory O_1X_1 at a speed of v. The symbol P is an arbitrary target in the imaging area. We define the vector pointing toward the target from the aperture center as the radar's light of sight, which is related to the location of the target. The angle between the radar's light of sight and the Z-axis is the beam pointing pitch angle α. The angle between the projection of the radar's light of sight on the ground and the X-axis is the beam pointing azimuth angle θ. The range between the target and the APC at the aperture center is the reference slant range r of the target. In the $OXYZ$ Cartesian coordinate system, the coordinate of the target P is expressed as $\mathbf{P}[x, y, z] = [r\sin\alpha\cos\theta, r\sin\alpha\sin\theta, 0]$. At instantaneous radar slow time t_m, the coordinate of the transmitter and receiver is defined as $\mathbf{R_T}[vt_m, 0, h]$ and $\mathbf{R_{Xi}}[vt_m, y_i, h]$, respectively. y_i denotes the Y-coordinate of the ith receiver array element. Then, the instantaneous two-way slant range of the target P can be expressed as

$$R(t_m, x, y, y_i) = R_{\mathrm{Tx}}(t_m, x, y) + R_{\mathrm{Rx}}(t_m, x, y, y_i) \tag{1}$$

where $R_{\mathrm{Tx}}(t_m, x, y)$ and $R_{\mathrm{Rx}}(t_m, x, y, y_i)$ denote the receiver instantaneous slant range and the transmitter instantaneous slant range, respectively

$$\begin{aligned} R_{\mathrm{Tx}}(t_m, x, y) &= \|\mathbf{R_T} - \mathbf{P}\| \\ &= \sqrt{(vt_m - x)^2 + y^2 + h^2} \end{aligned} \tag{2}$$

$$\begin{aligned} R_{\mathrm{Rx}}(t_m, x, y, y_i) &= \|\mathbf{R_{Xi}} - \mathbf{P}\| \\ &= \sqrt{(vt_m - x)^2 + (y - y_i)^2 + h^2}. \end{aligned} \tag{3}$$

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Assume that the FLMC-SAR transmits linear frequency-modulated (LFM) pulse signals, which is one of the most widely used signals in the SAR imaging system

$$s(\tau) = \text{rect}\left(\frac{\tau}{T_p}\right) \cdot \exp\left[j2\pi\left(f_c\tau + \frac{\gamma}{2}\tau^2\right)\right] \tag{4}$$

where rect[] is the rectangular window function, τ denotes the radar fast time, T_P denotes the time width of the LFM pulse, f_c denotes the carrier frequency of the radar system, and γ denotes the chirp rate of LFM signal.

Through downconversion, the baseband received signal of the target P can be expressed as

$$\begin{aligned} s_0(\tau, t_m, x, y, y_i) = {} & A_P \cdot w_r\left(\tau - \frac{R(t_m, x, y, y_i)}{c}\right) \\ & \cdot \exp\left[\mathrm{j}\pi\gamma\left(\tau - \frac{R(t_m, x, y, y_i)}{\mathrm{c}}\right)^2\right] \\ & \cdot \exp\left[-j\frac{2\pi}{\lambda}R(t_m, x, y, y_i)\right] \end{aligned} \tag{5}$$

where A_P denotes the reflectivity coefficient of the target P, w_r denotes the window function in range, and c denotes the light speed.

B. FLMC-SAR Imaging

In order to better present our FLMC-SAR image resolution enhancement method, the forward-looking area is preliminarily distinguished. Let us first introduce the FLMC-SAR imaging.

In order to achieve a range of high resolution, the signal is subjected to pulse compression processing

$$\begin{aligned} & s_1(\tau, t_m, x, y, y_i) \\ & \quad = \text{IFT}\{\text{FT}[s_0(\tau, t_m, x, y, y_i)]\text{FT}[s_{\text{ref}}(\tau)]\} \\ & \quad = A_P \cdot \text{sinc}\left[B\left(\tau - \frac{R(t_m, x, y, y_i)}{c}\right)\right] \\ & \qquad \cdot \exp\left[-j\frac{2\pi}{\lambda}R(t_m, x, y, y_i)\right] \end{aligned} \tag{6}$$

where B denotes the bandwidth of LFM pulse, FT[·] and IFT[·] are, respectively, the Fourier transform operator and inverse Fourier transform operator, sinc[·] is the sampling function, and $s_{\text{ref}}[\cdot]$ is the matching filter function, which can be described by

$$s_{\text{ref}}(\tau) = \text{rect}\left(\frac{\tau}{T_p}\right)\exp(-j\pi\gamma\tau^2). \tag{7}$$

To achieve high azimuth resolution, the range cell migration correction (RCMC) is performed. The detailed RCMC processes can be found in Appendix A.

After RCMC, the signal can be expressed as

$$\begin{aligned} & s_1(\tau, t_m, x, y, y_i) \\ & \quad = \text{IFT}\{\text{FT}[s_0(\tau, t_m, x, y, y_i)] \cdot \text{FT}[s_{\text{ref}}(\tau)]\} \\ & \quad = A_P \cdot \text{sinc}\left[B\left(\tau - \frac{2(r_P - r_0)}{c}\right)\right] \\ & \qquad \cdot \exp\left[-j\frac{2\pi}{\lambda}[R(t_m, x, y, y_i) - R(t_m, x_0, y_0, y_i)]\right] \end{aligned} \tag{8}$$

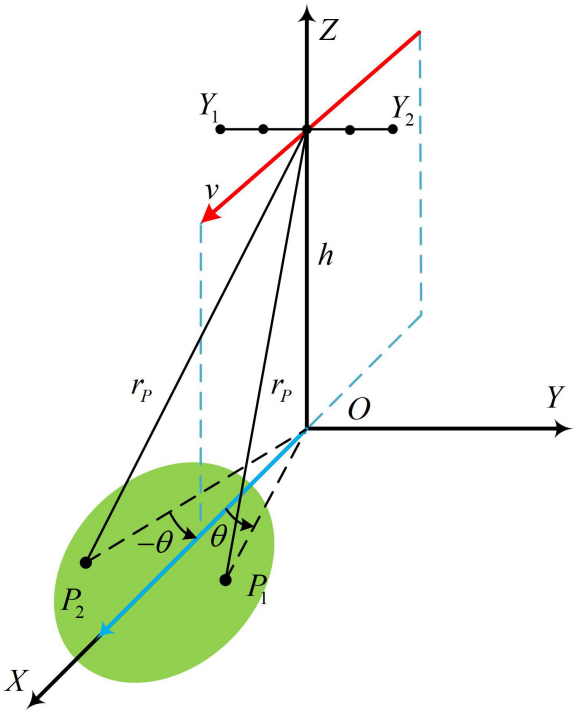

Fig. 1. Geometry model with two symmetric targets.

where r_P is the reference slant range of the target P, r_0 is the reference slant range of the scene center, x_0 is the x-coordinate of the scene center, and y_0 is the y-coordinate of the scene center.

C. Doppler Ambiguity in FLMC-SAR System

In Fig. 1, there are two targets P_1 and P_2 in the imaging area, with the same reference slant range r_P, but opposite azimuth angles. P_1 and P_2 are symmetrical about the X-axis (trajectory), and their coordinates are $\mathbf{P_1}[x, -y, z] = [r_P \sin\alpha\cos\theta, -r_P \sin\alpha \sin\theta, 0]$ and $\mathbf{P}_2[x, y, z] = [r_P \sin\alpha\cos\theta, r_P \sin\alpha \sin\theta, 0]$, respectively.

Equation (8) indicates that after range pulse compression and RCMC, the target signal will be focused on the range unit according to the reference range r_P. Therefore, the signals of targets P_1 and P_2 will be focused in the same range unit, which is expressed as follows:

$$\begin{aligned} & s_2(\tau, t_m, x, y, y_i) \\ & \quad = s_1(\tau, t_m, x, -y, y_i) + s_1(\tau, t_m, x, y, y_i) \\ & \quad = A_{P1}\text{sinc}\left[B\left(\tau - \frac{2(r_P - r_0)}{c}\right)\right] \\ & \qquad \cdot \exp\left\{-j\frac{2\pi}{\lambda}[R(t_m, x, -y, y_i) - R(t_m, x_0, y_0, y_i)]\right\} \\ & \qquad + A_{P2}\text{sinc}\left[B\left(\tau - \frac{2(r_P - r_0)}{c}\right)\right] \\ & \qquad \cdot \exp\left\{-j\frac{2\pi}{\lambda}[R(t_m, x, y, y_i) - R(t_m, x_0, y_0, y_i)]\right\} \\ & \quad = \text{sinc}\left[B\left(\tau - \frac{2(r_P - r_0)}{c}\right)\right] \end{aligned}$$

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$$\cdot\left\{A_{P1}\cdot\exp\left\{-j\frac{2\pi}{\lambda}[R(t_m,x,-y,y_i)-R(t_m,x_0,y_0,y_i)]\right\}\right.$$
$$\left.+A_{P2}\cdot\exp\left\{-j\frac{2\pi}{\lambda}[R(t_m,x,y,y_i)-R(t_m,x_0,y_0,y_i)]\right\}\right\}. \tag{9}$$

The first term in (9) $\mathrm{sinc}[B(\tau-(2(r_P-r_0)/c))]$ is the range envelope of the signals, which determines the range unit that the target will be focused on after range pulse compression and RCMC. The second term in (9) includes the reflection coefficient A_{P1} and azimuth phase $\exp[-j(2\pi/\lambda)R(t_m,x,-y,y_i)]$ of the target P_1, where the azimuth phase represents the Doppler frequency of the target P_1 and determines which azimuth unit the target P_1 is focused on. Similarly, the third term in (9) $\exp[-j(2\pi/\lambda)R(t_m,x,y,y_i)]$ represents the Doppler frequency of the target P_2 and determines which azimuth unit the target P_2 is focused on.

Calculate the instantaneous two-way slant range difference between targets P_1 and P_2, and the Taylor expansion is expressed as follows:

$$R(t_m,x,-y,y_i)-R(t_m,x,y,y_i)\approx A+Bt_m+Ct_m^2. \tag{10}$$

For simplicity, let

$$R_1=\sqrt{(x)^2+(y+y_i)^2+h^2} \tag{11}$$
$$R_2=\sqrt{(x)^2+(y-y_i)^2+h^2}. \tag{12}$$

Then,

$$A=R_1-R_2 \tag{13}$$
$$B=-\frac{\upsilon x}{R_1}+\frac{\upsilon x}{R_2} \tag{14}$$
$$C=\left(\frac{\upsilon^2}{R_1}-\frac{\upsilon^2}{R_2}\right)-\left(\frac{\upsilon^3x^2}{R_1^3}-\frac{\upsilon^3x^2}{R_2^3}\right). \tag{15}$$

The antenna element spacing is on the wavelength scale, and therefore, the instantaneous two-way slant range difference between targets P_1 and P_2 is much smaller than the wavelength. Finally, the imaging results of symmetrical targets P_1 and P_2 will be focused in the same Doppler unit, which results in left–right Doppler ambiguity in FLMC-SAR imaging.

It is worth noting that the Doppler ambiguity is 2-D, that is, the imaging results of at most two targets are aliased in the same imaging unit.

Applying the inverse Fourier transform to (9) over t_m, we can obtain the ambiguous FLMC-SAR imaging results, which can be expressed as

$$\begin{aligned}&s_3(\tau,t_m,x,y,y_i)\\&=\mathrm{FT}[s_2(\tau,t_m,x,y,y_i)]\\&=A_{P1}\cdot\mathrm{sinc}\left[B\left(\tau-\frac{2(r_P-r_0)}{c}\right)\right]\\&\quad\cdot\mathrm{sinc}\{B_a[f-\sin\alpha\upsilon(1-\cos\theta)]\}\\&\quad\cdot\left\{A_{P1}\cdot\exp\left\{-j\frac{2\pi}{\lambda}[R(0,x,-y,y_i)-R(0,x_0,y_0,y_i)]\right\}\right.\\&\quad\left.+A_{P2}\cdot\exp\left\{-j\frac{2\pi}{\lambda}[R(0,x,y,y_i)-R(0,x_0,y_0,y_i)]\right\}\right\}\end{aligned} \tag{16}$$

where the first term is the range envelope of the signals, the second term is the azimuth envelope of the signals, and the third term is the phase difference between array steering vectors. After simplification, the FLMC-SAR imaging results of the kth channel [see (16)] can be expressed as follows:

$$\begin{aligned}&s_3(\tau,t_m,x,y,k)\\&=\mathrm{FT}[s_2(\tau,t_m,x,y,y_i)]\\&=A_{P1}\cdot\mathrm{sinc}\left[B\left(\tau-\frac{2(r_P-r_0)}{c}\right)\right]\\&\quad\cdot\mathrm{sinc}\{B_a[f-\sin\alpha\upsilon(1-\cos\theta)]\}\\&\quad\cdot\left\{A_{P1}\cdot\exp\left[-j\frac{2\pi}{\lambda}kd\sin\alpha(-\sin\theta)\right]\right.\\&\quad\left.+A_{P2}\cdot\exp\left[-j\frac{2\pi}{\lambda}kd\sin\alpha\sin\theta\right]\right\}.\end{aligned} \tag{17}$$

D. FLMC-SAR Space–Time Observation Model

The array's degrees of- freedom are critical for both FLMC-SAR image resolution enhancement and Doppler ambiguity resolving. Regarding the array's degrees of freedom as a key factor, we established the following FLMC-SAR space–time observation model:

$$\mathbf{S}(x,y)_{K\times1}=\mathbf{H}_{K\times N}\mathbf{A}(x,y)_{N\times1}+\varepsilon_{K\times1}. \tag{18}$$

The space–time observation model is based on the preliminary FLMC-SAR images. The observations of each channel at the target coordinate (x,y) are the imaging results at each channel, which are expressed with a $K\times1$ matrix, denoted by $\mathbf{S}(x,y)_{K\times1}$. K is the number of channels. $\mathbf{A}(x,y)_{N\times1}$ is the resolution-enhanced FLMC-SAR image at the target coordinate (x,y). N is the number of imaging units. $\varepsilon_{K\times1}$ is the additive measurement noise term. $\mathbf{H}_{K\times N}$ is a $K\times N$ matrix whose columns represent the steering vectors of the targets in the imaging area, which is expressed as follows:

$$\mathbf{H}_{K\times N}=[h_1,h_2,\ldots,h_N] \tag{19}$$

where h_i denotes the steering vector of the ith imaging unit, which is expressed as follows:

$$h_i=\begin{bmatrix}\exp\left(-j\frac{2\pi}{\lambda}d\sin\alpha_i\sin\theta_i\right)\\\exp\left(-j\frac{2\pi}{\lambda}2d\sin\alpha_i\sin\theta_i\right)\\\cdots\\\exp\left(-j\frac{2\pi}{\lambda}Kd\sin\alpha_i\sin\theta_i\right)\end{bmatrix}_{K\times1} \tag{20}$$

where θ_i and α_i are the azimuth angle and pitch angle of the ith imaging unit, respectively. It can be seen from (20) that more array channels will reduce the cross correlations of the steering vectors. Also, small cross correlations usually can guarantee the accuracy of sparse reconstruction. Therefore, the number of channels is an important factor affecting the spatial freedom of the FLMC-SAR system, which is closely related to the resolution enhancement performance.

In practice, due to the Doppler ambiguity, the imaging results of the targets $P_1(x,-y)$ and $P_2(x,y)$ are aliased in

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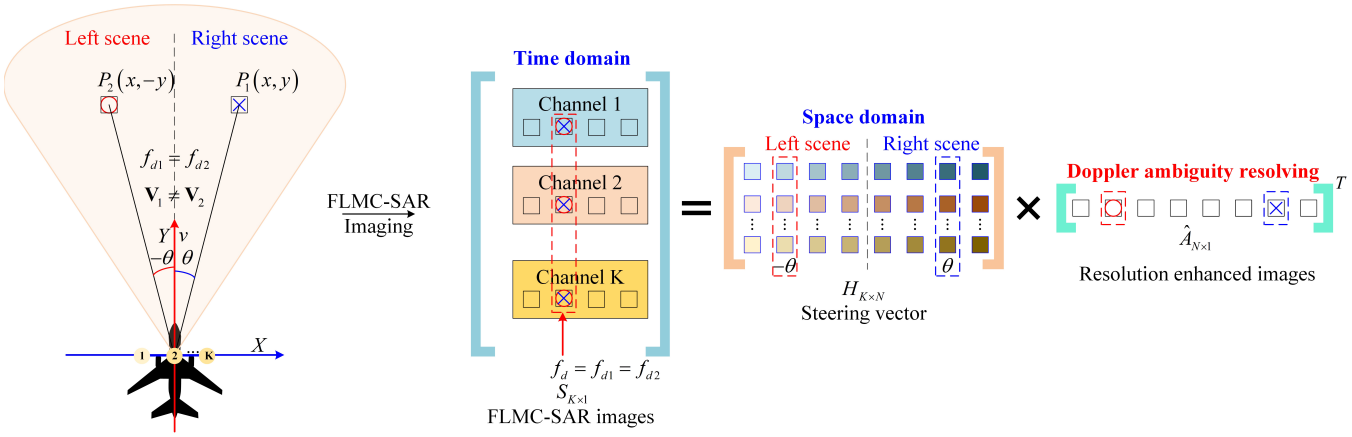

Fig. 2. Doppler ambiguity resolving.

the same imaging unit. As a result, the measurement equation for FLMC-SAR signals can be written as

$$\begin{aligned}&\mathbf{S}_4(x,y)_{K\times1}\\&\quad=\mathbf{S}_4(x,-y)_{K\times1}\\&\quad=\mathbf{S}(x,y)_{K\times1}+\mathbf{S}(x,-y)_{K\times1}\\&\quad=\mathbf{H}_{K\times N}\mathbf{A}(x,y)_{N\times1}+\mathbf{H}_{K\times N}\mathbf{A}(x,-y)_{N\times1}+\varepsilon_{K\times1}\\&\quad=\mathbf{H}_{K\times N}\hat{\mathbf{A}}(x,y)_{N\times1}+\varepsilon_{K\times1}\end{aligned}\tag{21}$$

where $\hat{\mathbf{A}}(x,y)_{N\times1}$ is the resolution-enhanced FLMC-SAR image at the target coordinate $(x,-y)$ and (x,y).

Compared with the traditional CS algorithm, the proposed space–time observation model has the following two advantages.

First, the array's degrees of freedom are regarded as an important factor in this space–time observation model. The columns of the matrix $\mathbf{H}_{K\times N}$ represent the steering vectors of the target in different imaging units, which is significant for Doppler ambiguity resolving. As shown in Fig. 2, the array antenna is distributed on the X-axis, and the velocity direction of the radar platform is the Y-axis. The normal direction of the array antenna is the Y-axis. There are two targets in the imaging area, which are symmetrical about the velocity direction of the radar platform. The coordinates of the symmetric targets P_1 and P_2 are (x,y) and $(x,-y)$, respectively. The azimuth angle of the symmetric targets P_1 and P_2 are θ and $-\theta$, respectively. The Doppler frequencies of P_1 and P_2 are

$$f_{d1}=\frac{2v\cos(\theta)}{\lambda}\tag{22}$$

$$f_{d2}=\frac{2v\cos(-\theta)}{\lambda}.\tag{23}$$

The steering vectors of P_1 and P_2 are

$$\mathbf{V}_1=\begin{bmatrix}\exp\left(-j\frac{2\pi}{\lambda}d\sin\alpha\sin\theta\right)\\\exp\left(-j\frac{2\pi}{\lambda}2d\sin\alpha\sin\theta\right)\\\cdots\\\exp\left(-j\frac{2\pi}{\lambda}Kd\sin\alpha\sin\theta\right)\end{bmatrix}_{K\times1}\tag{24}$$

$$\mathbf{V}_2=\begin{bmatrix}\exp\left(-j\frac{2\pi}{\lambda}d\sin\alpha\sin(-\theta)\right)\\\exp\left(-j\frac{2\pi}{\lambda}2d\sin\alpha\sin(-\theta)\right)\\\cdots\\\exp\left(-j\frac{2\pi}{\lambda}Kd\sin\alpha\sin(-\theta)\right)\end{bmatrix}_{K\times1}.\tag{25}$$

It can be seen that the symmetric targets P_1 and P_2 have the same Doppler frequency, but the steering vectors of them are opposite

$$f_d=f_{d1}=f_{d2}=\frac{2v\cos\theta}{\lambda}\tag{26}$$

$$\mathbf{V}_1=\text{conj}(\mathbf{V}_2)\tag{27}$$

where $\text{conj}(x)$ is the conjugate of x. In the processing framework of this article, time-domain SAR imaging is first performed, which achieves target azimuth resolution through the difference in Doppler frequency. The symmetric targets P_1 and P_2 with the same Doppler frequency will be focused in the same azimuth resolution unit, resulting in Doppler ambiguity in FLMC-SAR imaging. Based on SAR imaging, we construct the observation matrix, whose columns represent the steering vector of the targets with different azimuth angles, as shown in (19). The difference between the spatial steering vectors of the symmetrical targets P_1 and P_2 can be used to separate left scenes from right scenes to achieve Doppler ambiguity resolving.

Second, the Doppler ambiguity is 2-D, so the resolution-enhanced image $\hat{\mathbf{A}}(x,y)_{N\times1}$ contains up to two targets with different azimuth unit, which means that $\hat{\mathbf{A}}(x,y)_{N\times1}$ is highly sparse. The FLMC-SAR signal model with exploiting space–time sparsity is shown in Fig. 3. Because the SAR imaging scene does not have strong spatial sparsity, the traditional observation model limits the accuracy of reconstructed images. In the proposed method, an FLMC space–time observation model is established through space–time cascade processing. The range–azimuth decoupling is first realized in the FLMC-SAR imaging process, so the reconstruction can be carried out on a range-by-range bin basis. Meanwhile, coarse azimuth imaging results can ensure that reconstruction can

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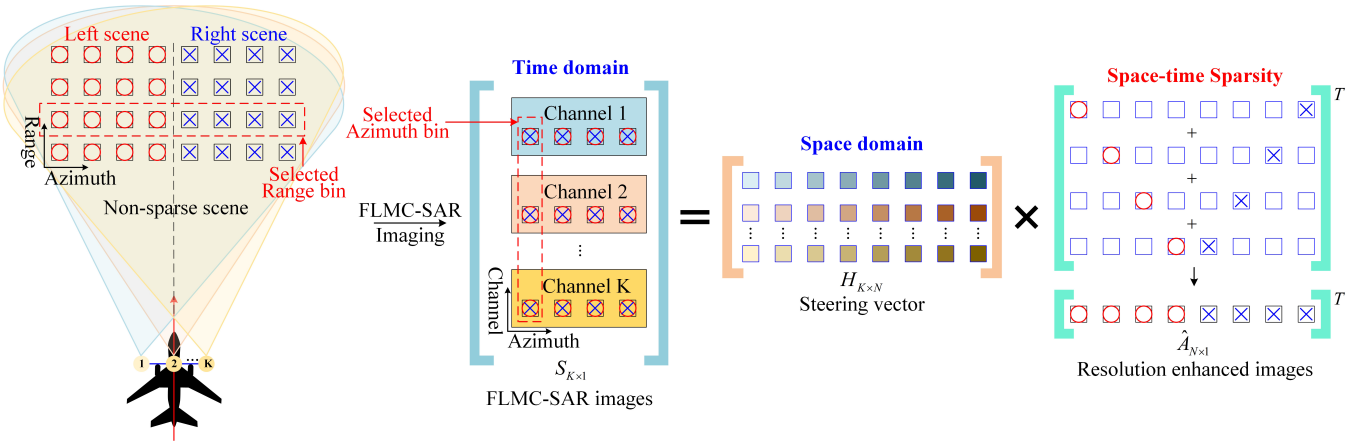

Fig. 3. FLMC-SAR signal model with exploiting space–time sparsity.

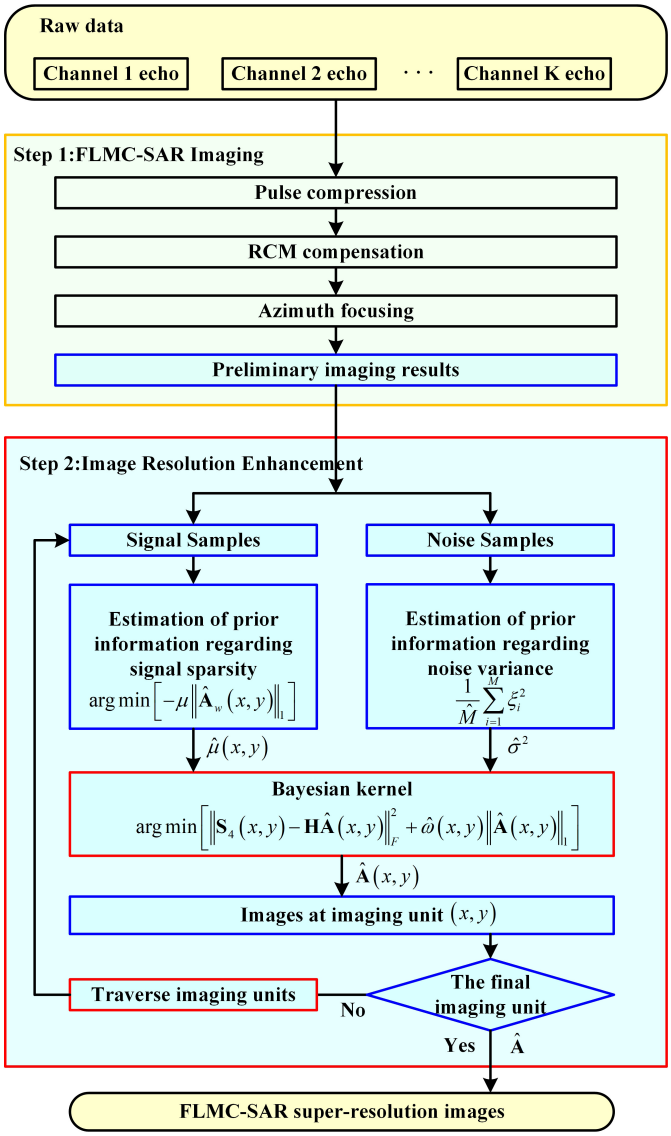

Fig. 4. Flowchart of the proposed method.

be performed on an azimuth-by-azimuth bin basis. With the selected range bin and the selected azimuth bin, the multi-channel signals of different imaging units can be separated in the image domain. Therefore, the separated signal of each imaging unit has strong space–time sparsity. On this basis, we can achieve the high-precision sparse reconstruction of the signal in each imaging unit. Finally, the resolution enhancement can be achieved by accumulating the sparse reconstruction results of each imaging unit.

TABLE I
SIMULATION PARAMETERS

Parameter	Value	Parameter	Value
Carrier frequency	30GHz	Platform velocity	50m/s
Bandwidth	60MHz	Platform height	4000m
PRF	5000Hz	Number of array elements	5
Synthetic aperture time	2s	Real array length	0.16m

III. FLMC-SAR Image Resolution Enhancement

A. Bayesian Kernel

Based on the proposed observations, we propose a quadratic regularization-based method for FLMC-SAR image resolution enhancement through the Bayesian kernel.

Assume that the measurement noise in the space–time observation model is independent identically distributed Gaussian noise. If σ^2 is the variance of the measurement noise, then its probability density function (pdf) is expressed as

$$P(\varepsilon)=\frac{1}{\left(\sqrt{2\pi}\sigma\right)^{\hat{N}}}\exp\left(-\frac{\|\varepsilon\|_F^2}{2\sigma^2}\right) \tag{28}$$

where $\|\cdot\|_F$ is the Frobenius norm of the matrix. Then, the likelihood function of the observation is expressed as

$$P(\mathbf{S}_4(x,y)|\hat{\mathbf{A}}(x,y);\sigma)=\frac{1}{\left(\sqrt{2\pi}\sigma\right)^{\hat{N}}}\exp\left(-\frac{\left\|\mathbf{S}_4(x,y)-\mathbf{H}\hat{\mathbf{A}}(x,y)\right\|_F^2}{2\sigma^2}\right). \tag{29}$$

A widely used sparseness prior is the Laplace density function [12], [13]. The prior information form of signal

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TABLE II
IRW, ISLR, AND PSLR OF POINT TARGETS

Parameter	Azimuth angle	Reference slant range	Proposed method			Beamforming		
			IRW/m	ISLR/dB	PSLR/dB	IRW/m	ISLR/dB	PSLR/dB
Target 1	-3.2degree	Rs1	6.54	-11.08	-17.65	6.55	-9.88	-12.37
Target 2	0.4degree	Rs2	8.16	-10.09	-13.38	37.57	-14.76	-10.63
Target 3	4.0degree	Rs3	6.55	-9.78	-12.42	6.56	-11.35	-12.61

sparsity is given by

$$P(\hat{\mathbf{A}}(x,y);\mu)=\left(\frac{\mu}{2}\right)^{\hat{N}}\exp(-\mu\|\hat{\mathbf{A}}(x,y)\|_1) \tag{30}$$

where μ is the scalar parameter related to the prior information and $\|\cdot\|_1$ is the ℓ_1 norm of the matrix.

According to the Bayesian principle, the objective function of the maximum a posteriori (MAP) estimator using the prior information of signal sparsity can be obtained

$$\begin{aligned}\hat{\mathbf{A}}(x,y)&=\arg\max[P(\hat{\mathbf{A}}(x,y)|\mathbf{S}_4(x,y);\sigma)]\\&=\arg\max[P(\mathbf{S}_4(x,y)|\hat{\mathbf{A}}(x,y);\sigma)\cdot P(\hat{\mathbf{A}}(x,y);\mu)].\end{aligned} \tag{31}$$

Taking the logarithmic, (31) can be rewritten as

$$\begin{aligned}&\hat{\mathbf{A}}(x,y)\\&=\arg\max\{\log[P(\mathbf{S}_4(x,y)|\hat{\mathbf{A}}(x,y);\sigma)]\\&\qquad+\log[P(\hat{\mathbf{A}}(x,y);\mu)]\}\\&=\arg\max\left[-\frac{\left\|\mathbf{S}_4(x,y)-\mathbf{H}\hat{\mathbf{A}}(x,y)\right\|_F^2}{2\sigma^2}-\mu\|\hat{\mathbf{A}}(x,y)\|_1\right]\\&=\arg\min[\|\mathbf{S}_4(x,y)-\mathbf{H}\hat{\mathbf{A}}(x,y)\|_F^2+\omega\|\hat{\mathbf{A}}(x,y)\|_1]\end{aligned} \tag{32}$$

where ω is the scalar parameter related to the prior information, such as the sparsity and noise level of the signal. The objective function consists of two terms. The first term in (32) is the data fidelity term of the estimator. The second terms in (32) incorporate the signal sparsity with prior information.

B. Parameterized Prior Statistics Estimator

On each imaging unit in the preliminary image, we perform the Bayesian kernel, which relies on parameterized prior information. The scattered fields of the distributed target usually have spatial-assembling characteristics, and the scattering centers of the same distinguishable target are often clustered into a small image area. Within this image area, the signals in the imaging units are used to characterize the sparsity of the same distinguishable target, which is related to the prior statistics of this image area. It means that the signal sparsity with prior information of the entire image area is related to the spatial characteristics of the targets.

The Laplace density function used to characterize the signal sparsity is rewritten as

$$P(\hat{\mathbf{A}}(x,y);\mu(x,y))=\left(\frac{\mu}{2}\right)^{\hat{N}}\exp(-\mu(x,y)\|\hat{\mathbf{A}}(x,y)\|_1). \tag{33}$$

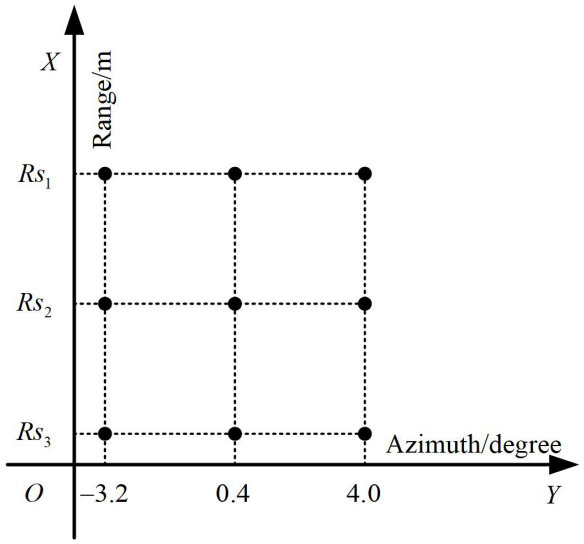

Fig. 5. Geometric distribution of point targets.

TABLE III
SYSTEM PARAMETERS

Parameter	Value	Parameter	Value
Carrier frequency	30GHz	Platform velocity	80m/s
Bandwidth	60MHz	Platform height	4000m
PRF	5000Hz	Number of array elements	5
Synthetic aperture time	1.2s	Real array length	0.16m

Note that the scalar parameter related to the prior information $\mu(x,y)$ in (33) is no longer constant but changes with the coordinates of the imaging unit (x,y), which characterizes the spatial characteristics of the targets. The preliminary imaging results of FLMC-SAR are the low-resolution samples of the image area, which are sufficient to estimate the sparsity of the image area. According to the preliminary imaging results of FLMC-SAR, the least square estimator is developed to estimate the statistics of noise and target signal so as to determine the parameters of the prior information.

When processing one imaging unit, we selected several adjacent imaging units as samples for parameterized prior information estimation. Then, the MAP estimator is given as

$$\begin{aligned}\hat{\mu}(x,y)&=\arg\max P(\hat{\mathbf{A}}_w(x,y);\mu(x,y)|a_1,a_2,\ldots,a_M)\\&=\arg\max[-\mu\|\hat{\mathbf{A}}_w(x,y)\|_1]\\&=\arg\min[-\mu\|\hat{\mathbf{A}}_w(x,y)\|_1]\end{aligned} \tag{34}$$

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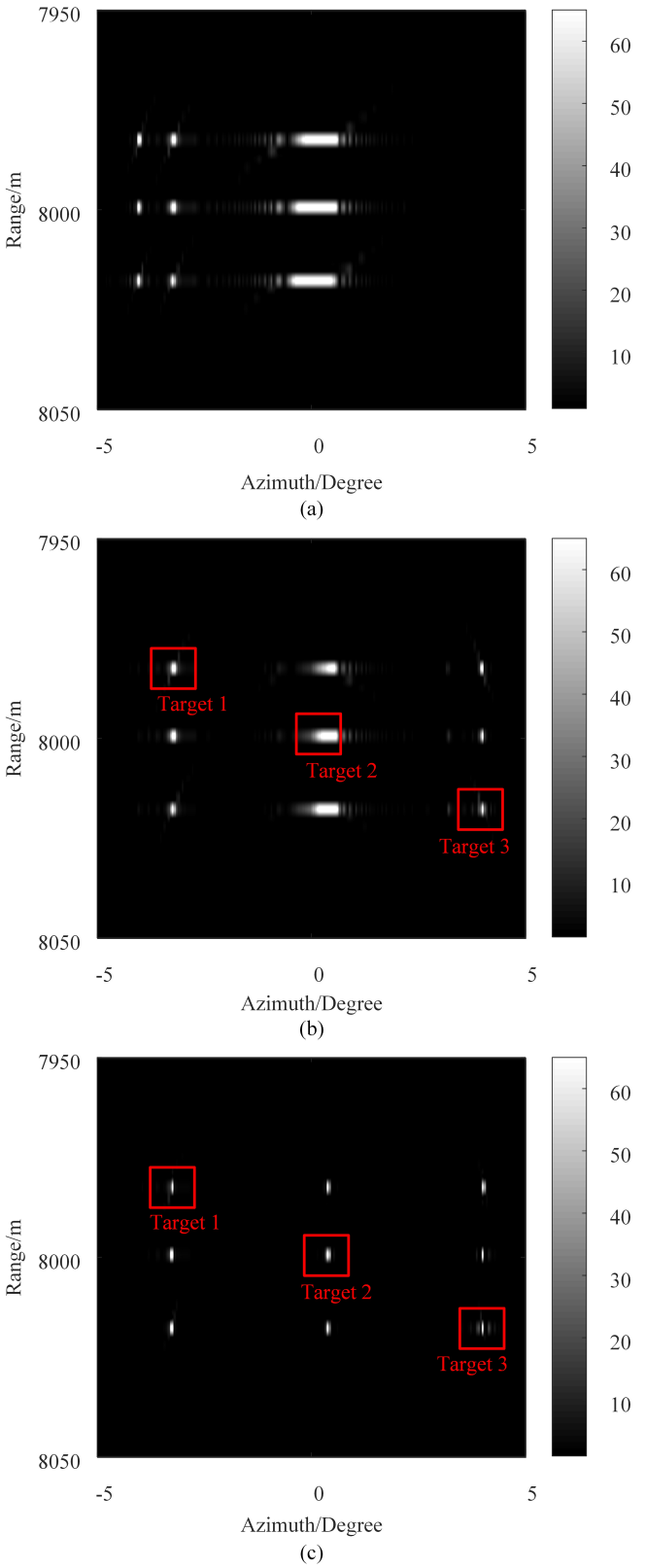

Fig. 6. Imaging results of the point targets. (a) SAR imaging result. (b) Imaging result via beamforming. (c) Resolution-enhanced images obtained by the proposed method.

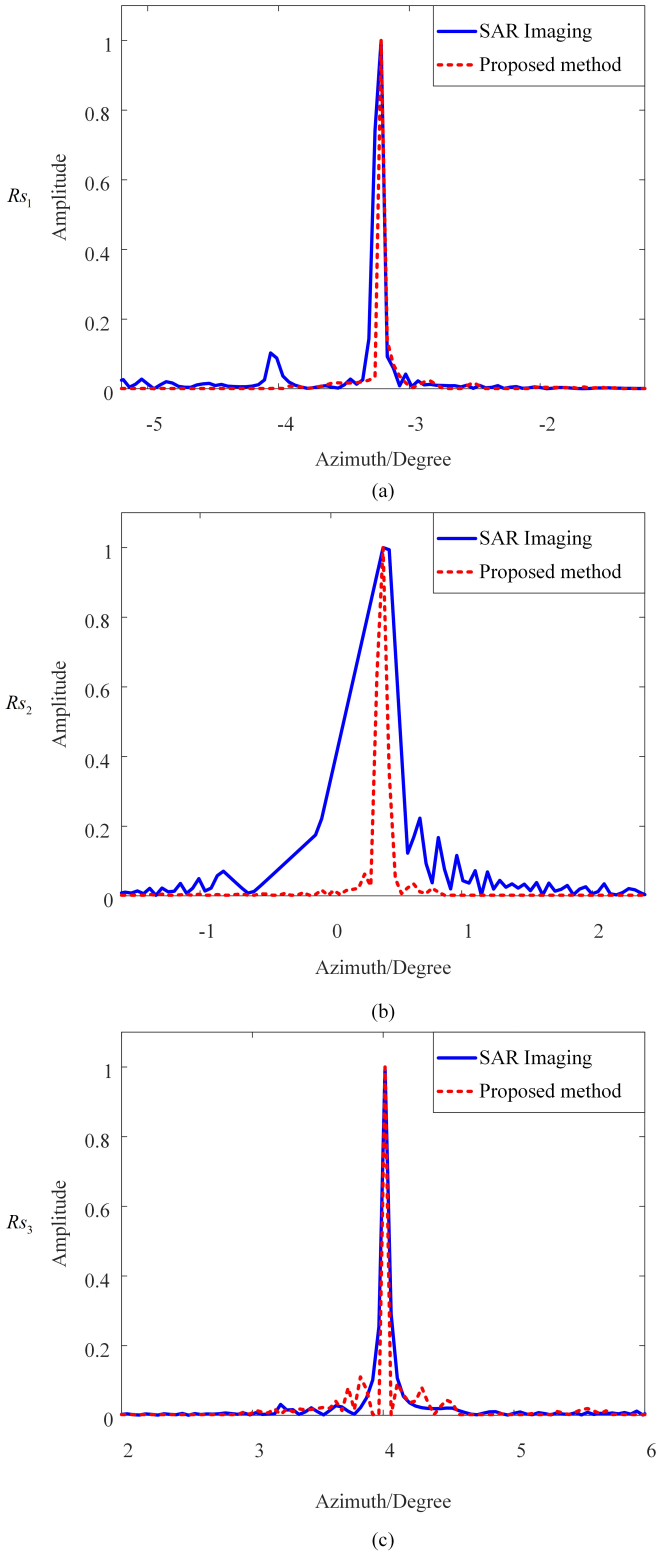

Fig. 7. Azimuth pulse response function of point targets processed by SAR imaging and the proposed method. (a) Azimuth pulse response function of Target 1. (b) Azimuth pulse response function of Target 2. (c) Azimuth pulse response function of Target 3.

where $\hat{\mathbf{A}}_w(x, y)$ is the signal samples in the preliminary images corresponding to the selected imaging units, which can be expressed as follows:

$$\hat{\mathbf{A}}_w(x, y) = [a_1, a_2, \ldots, a_M]. \tag{35}$$

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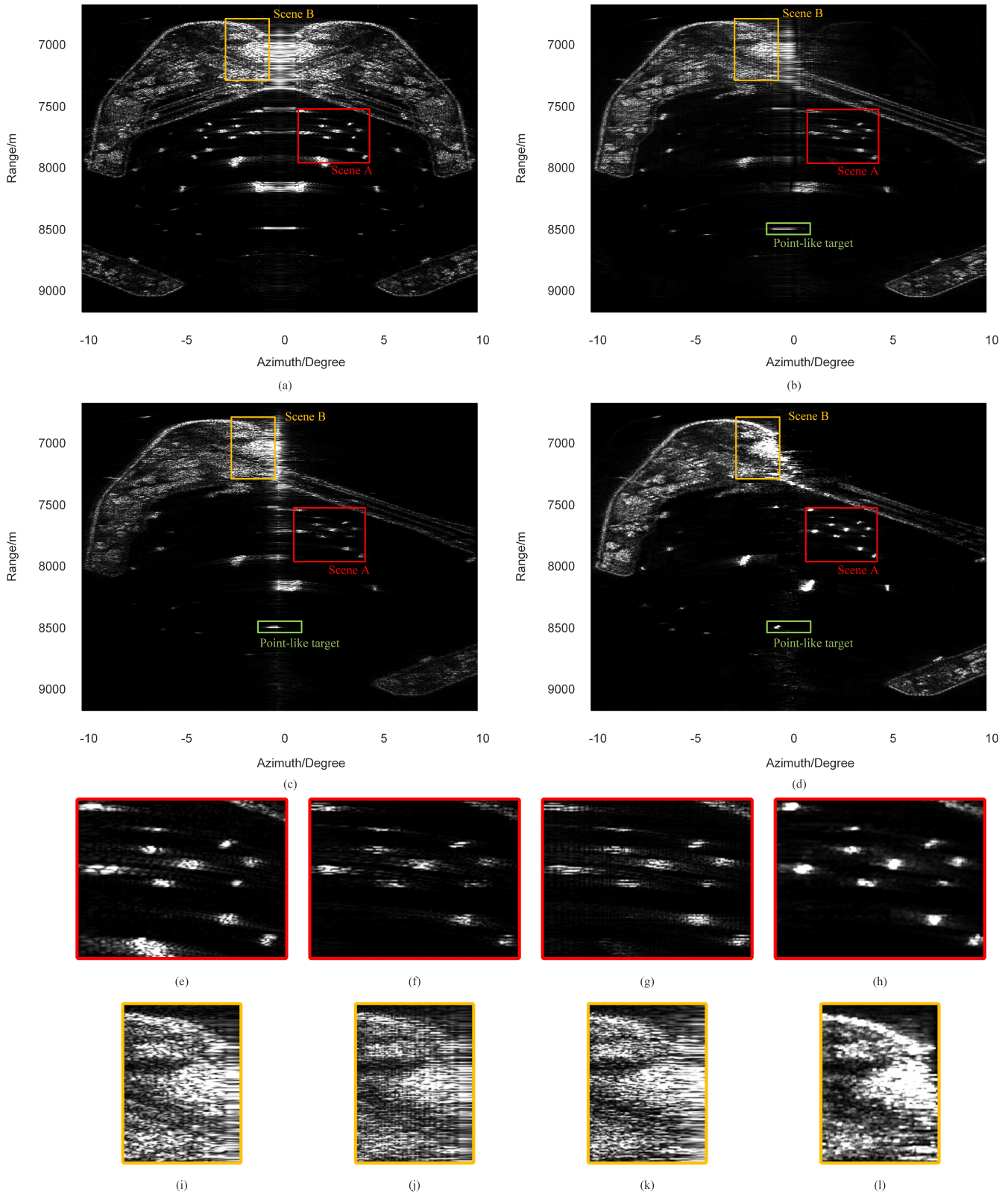

Fig. 8. Imaging results of the surface target. (a) SAR imaging result. (b) Imaging result via beamforming. (c) Imaging result via SBL. (d) Resolution-enhanced images obtained by the proposed method. (e) Scene A in SAR imaging result. (f) Scene A in imaging result via beamforming. (g) Scene A in imaging result via SBL. (h) Scene A in resolution-enhanced images obtained by the proposed method. (i) Scene B in SAR imaging result. (j) Scene B in imaging result via beamforming. (k) Scene B in imaging result via SBL. (l) Scene B in resolution-enhanced images obtained by the proposed method.

There are M selected samples, and a_i demotes the ith sample. The parameterized prior information characterizing the spatial structure of targets is obtained, and then, we perform noise variance estimation.

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According to (9), we know that the Doppler frequency of the target determines which azimuth unit the target focuses on. In SAR images, targets are generally stationary or low-speed targets, and the corresponding Doppler frequency is low. Therefore, targets will focus on an imaging area, and the maximum Doppler frequency in this area is determined by the geometry configuration of the system. The imaging area beyond the maximum Doppler frequency contains only noise. These noise samples can be used to estimate the noise variance.

The noise is independent identically distributed Gaussian noise. The MAP estimator is given as

$$\hat{\sigma}^2 = \frac{1}{\hat{M}} \sum_{i=1}^{\hat{M}} \xi_i^2. \tag{36}$$

There are $\hat{M}$ selected noise samples, and ξ_i^2 demotes the variance of the ith noise sample.

Substituting the above parameterized prior information into (32), we can obtain the following Bayesian kernel:

$$\begin{aligned}\hat{\mathbf{A}}(x,y) &= \arg\max\{\log[P(\mathbf{S}_4(x,y)|\hat{\mathbf{A}}(x,y);\hat{\sigma})] \\ &\quad + \log[P(\hat{\mathbf{A}}(x,y);\hat{\mu}(x,y))]\} \\ &= \arg\max\left[-\frac{\left\|\mathbf{S}_4(x,y)-\mathbf{H}\hat{\mathbf{A}}(x,y)\right\|_F^2}{2\hat{\sigma}^2}\right. \\ &\quad \left. -\hat{\mu}(x,y)\left\|\hat{\mathbf{A}}(x,y)\right\|_1\right] \\ &= \arg\min[\|\mathbf{S}_4(x,y)-\mathbf{H}\hat{\mathbf{A}}(x,y)\|_F^2 \\ &\quad + \hat{\omega}(x,y)\|\hat{\mathbf{A}}(x,y)\|_1]. \end{aligned} \tag{37}$$

Then, the iteration of quasi-Newton solver is expressed as

$$\hat{\mathbf{A}}(x,y)^{k+1} = \hat{\mathbf{A}}(x,y)^k - [\mathbf{H}(\hat{\mathbf{A}}(x,y)^k)]^{-1}\nabla f(\hat{\mathbf{A}}(x,y)) \tag{38}$$

where $\nabla f(\hat{\mathbf{A}}(x,y))$ is the conjugate gradient, $\mathbf{H}(\hat{\mathbf{A}}(x,y)^k)$ is the Hessian matrix, and $\hat{\mathbf{A}}(x,y)^k$ is the estimation if $\mathbf{A}(x,y)$ in the kth iteration. The derivation of the conjugate gradient is given in Appendix B.

C. Computational Complexity

Reconstructing all imaging units with (37), we get the FLMC-SAR resolution-enhanced image. The flowchart of the proposed method is shown in Fig. 4. The proposed method mainly consists of two steps.

1) FLMC-SAR Imaging: Assume that the number of range bins is M, the number of azimuth bins is N, and the number of channels is K. To achieve range pulse compression, we need to operate fast Fourier transform (FFT) $2KN$ times for the vector with the size of $M \times 1$. To achieve RCMC, we need to operate FFT $2KN$ times for the vector with the size of $M \times 1$ and to operate FFT $2KM$ times for the vector with the size of $N \times 1$. To achieve azimuth focusing, we need to operate FFT KM times for the vector with the size of $N \times 1$. The total computation of FLMC-SAR imaging is $O(KNM(\log M + \log N))$.

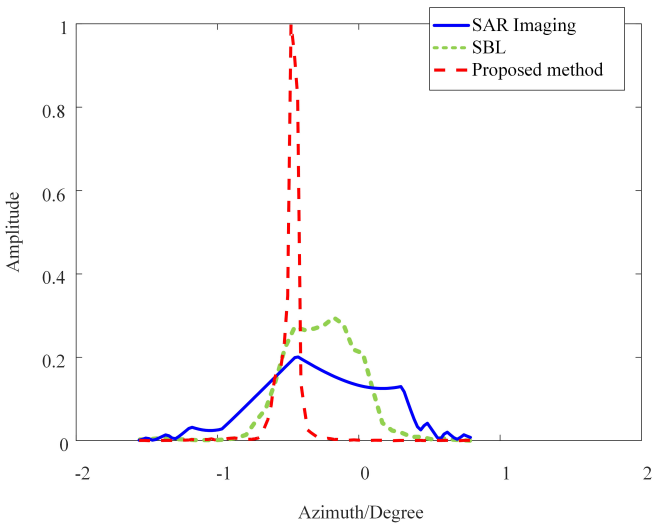

Fig. 9. Azimuth response function of point-like target.

2) Image Resolution Enhancement: Assume that the number of the iterations to achieve sparse reconstruction is N_{it}, and the samples of local estimator are $\hat{M}$. The computation of MN times MAP estimator with $\hat{M}$ samples is $O(MN\hat{M})$. To achieve MN times sparse reconstruction, we need to operate matrix inversion MNN_{it} times for the matrix with the size of $\hat{M}\hat{M}$. The total computation of image resolution enhancement is $O(MNN_{it}\hat{M}^3)$.

The total computation of the proposed method is of order $O(NM(K\log M + K\log N + N_{it}\hat{M}^3))$.

IV. Experiment and Discussion

To verify the effectiveness of the proposed algorithm, we conducted simulated and real-data experiments. There are three cases in this section, including point target simulation, surface target simulation, and real data experiment.

A. Point Target Simulation

First, the point target simulation is performed. The geometric distribution of point targets is shown in Fig. 5. There are 3×3 lattice targets evenly distributed on the ground. Here, the X-axis denotes the target reference slant range and the Y-axis denotes the target azimuth angle. In range bin R_{s1}, R_{s2}, and R_{s3}, three targets are evenly distributed with 3.6° azimuth angle spacing. The radar system is equipped with a single antenna as the transmitting antenna and a uniform linear array as the receiving antenna. The LFM pulse signal is transmitted by the radar system, and the relevant simulation parameters are shown in Table I.

Setting the signal-to-noise ratio (SNR) to 20 dB, we performed SAR imaging and image resolution enhancement. The results are shown in Fig. 6.

FLCM-SAR imaging encounters the Doppler ambiguity. In the FLMC-SAR imaging method, the Doppler ambiguity is expressed by mixing the imaging result of the scene on the right side of the trajectory with the imaging result on the left side. Fig. 6(a) shows the imaging results of the point targets

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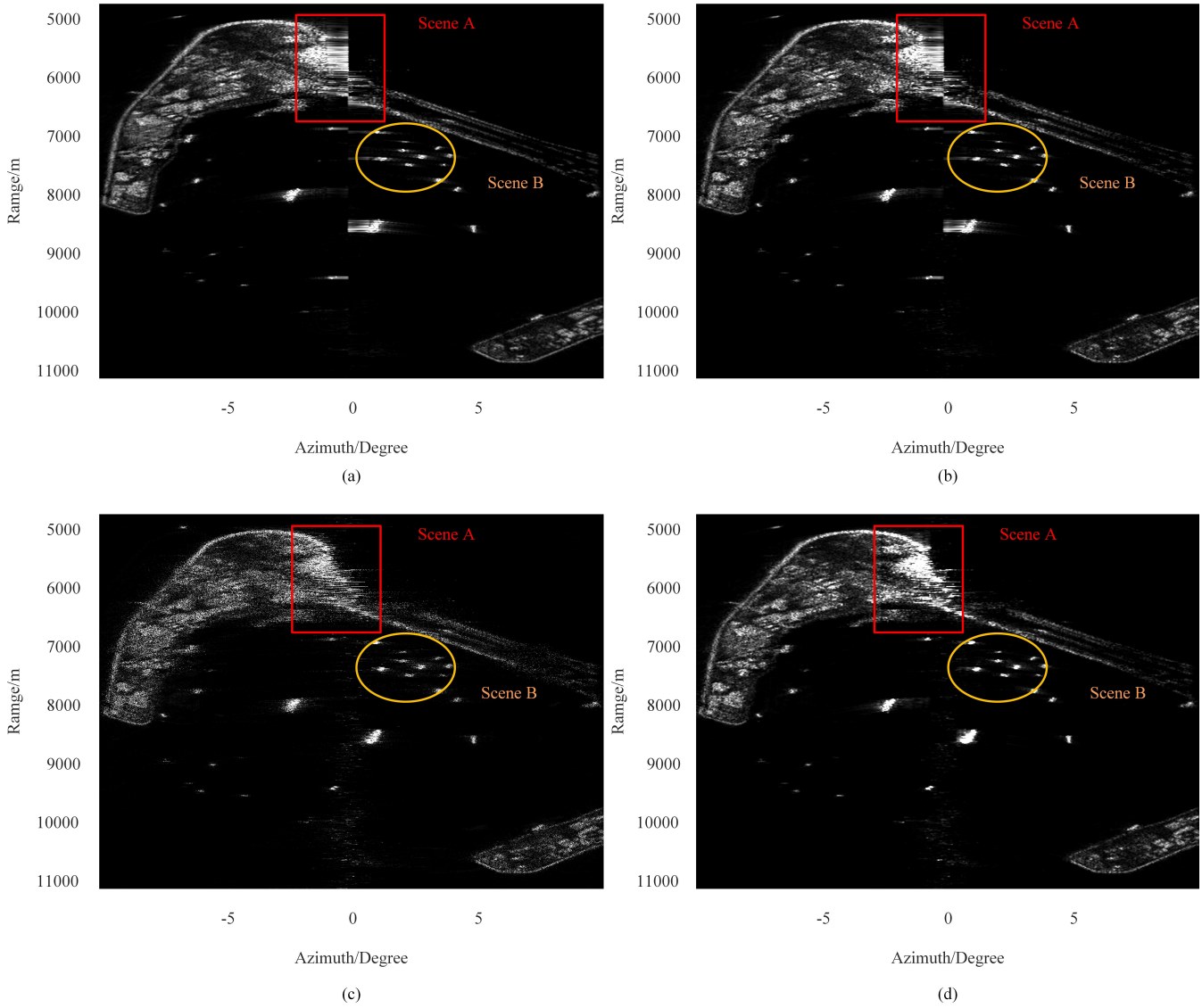

Fig. 10. Resolution-enhanced images of the surface target with different prior information parameters. (a) $\omega = 10$. (b) $\omega = 50$. (c) $\omega = 100$. (d) $\omega = \hat{\omega}$.

obtained by SAR imaging. The target with azimuth angle $-3.2°$ is on the left side of the trajectory, and the targets with azimuth angle 0.4° and 4° are on the right of the trajectory. From Fig. 6(a), it can be seen that the imaging results of the right scene are aliased with the imaging results of the left scene. Beamforming can solve the Doppler ambiguity [7], [8], and the image result without ambiguity is shown in Fig. 6(b). However, the target resolution in Fig. 6(b) is limited, especially in the area close to zero azimuth angle. Fig. 6(c) shows the resolution-enhanced images obtained by the proposed method. In the proposed method, the space–time observation model of FLMC-SAR is presented by analyzing joint FLMC-SAR imaging and Doppler ambiguity so that resolution-enhanced images without ambiguity can be obtained. There are three targets highlighted in Fig. 6(b) and (c), which have different rang bins and different azimuth bins. To show the effectiveness of the proposed algorithm, we also give the azimuth pulse response function of the point target highlighted in Fig. 6, as shown in Fig. 7. In Fig. 7, the amplitude of the azimuth pulse response function is normalized. We can see the improvement of the proposed method on the focusing for the targets, especially Target 2 in the area close to zero azimuth angle. Impulse response width (IRW), integral sidelobe ratio (ISLR), and peak sidelobe ratio (PSLR) are utilized to evaluate the point impulse responses, as shown in Table II. The experimental results show that the proposed method can enhance the resolution of the FLMC-SAR image and solve the Doppler ambiguity.

B. Surface Target Simulation

In this section, a simulation experiment of the surface target is performed. The relevant simulation parameters are shown in Table I. The simulation uses the SAR image obtained by real data as a reference image. Fig. 8(a) shows the imaging result obtained by the SAR imaging. It can be seen that the Doppler ambiguity makes the imaging result of the scene on the left side of the trajectory and that on the right side of the trajectory aliased together. According to the beamforming-based left–right Doppler ambiguity resolving method in [7]

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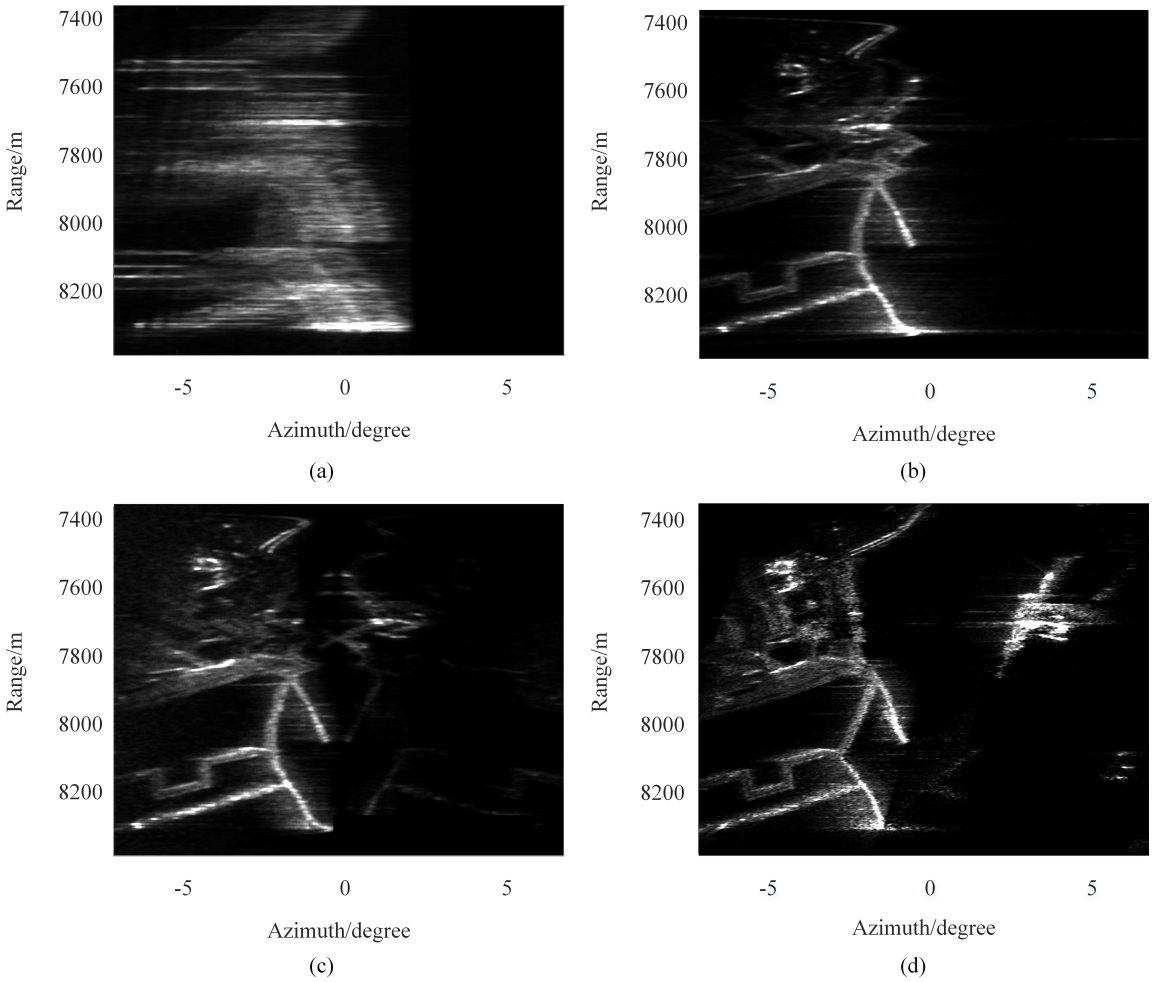

Fig. 11. Imaging results of real data. (a) Without motion compensation. (b) With motion compensation. (c) Imaging result via beamforming. (d) Resolution-enhanced images obtained by the proposed method.

and [8], we can get the FLMC-SAR image without ambiguity, as shown in Fig. 8(b). In the area close to the trajectory, that is, the area near zero azimuth angle, the resolution of the system is limited, which limits the imaging performance of FLMC-SAR. Based on the beamforming results, we employ sparse Bayesian learning (SBL) [39], [40], [41], [42] for resolution enhancement, as shown in Fig. 8(c). Analyzing joint FLMC-SAR imaging and Doppler ambiguity, a constrained optimization problem for FLMC-SAR image resolution enhancement is formulated, which combines the signal sparsity with prior information regarding the spatial structure of the target. The imaging result of the surface target via the proposed method is shown in Fig. 8(d). A scene with strong sparsity and another scene with poor sparsity are named Scenes A and B, respectively. Among them, the SBL algorithm has not undergone imaging processing, and the observation model is based on all the targets in the same range bin, so Scene B does not have strong sparsity under the modified model, which limits the reconstruction accuracy. Based on coarse imaging results, the proposed algorithm establishes a space–time observation model for each imaging unit. In this model, the signals separated by imaging have strong sparsity, so the proposed algorithm can achieve resolution enhancement for both strong sparse scenes and weak sparse scenes. A point-like target is highlighted by the green rectangle in Fig. 8(b)–(d), and its azimuth response function is shown in Fig. 9. Comparing the imaging results of different algorithms, the proposed method achieves well-focused imaging. The experimental results show that the resolution of FLMC-SAR images is enhanced via the proposed method, especially in the area near zero azimuth angle.

The signal sparsity is closely related to the prior information regarding the spatial structure of the target. Especially for the surface targets with complex spatial structures, the signal sparsity of targets in different scenes is different. In order to verify the influence of the prior information parameters on the reconstruction accuracy of targets with different spatial structures, we give the reconstruction results of the surface targets with different prior information parameters, as shown in Fig. 10. Two scenes close to the trajectory with different spatial structures are highlighted in Fig. 10, namely, Scenes A and B. In Fig. 10(a) and (b), the prior information parameters are $\omega = 10$ and $\omega = 50$, respectively. The sparsity of the corresponding imaging scene is not so strong. Scene B can

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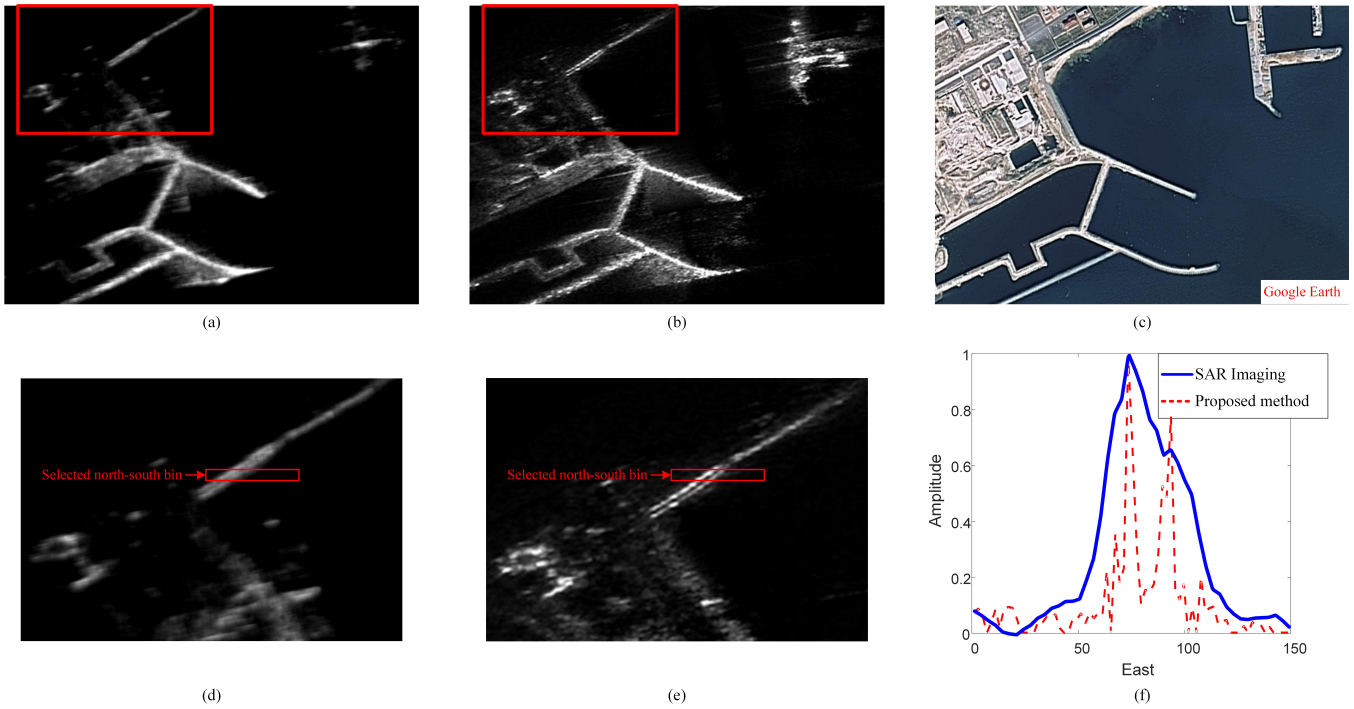

Fig. 12. Imaging results of real data in the geographic coordinate system. (a) Imaging results in the geographic coordinate system via beamforming. (b) Imaging results in the geographic coordinate system via the proposed method. (c) Satellite image of the imaging area. (d) Magnified images of imaging results via beamforming. (e) Magnified images of imaging results via the proposed method. (f) East–west profile of the two roads in the selected north–south bin.

be successfully reconstructed, but the reconstruction of Scene A fails. In Fig. 10(c), the prior information parameter is $\omega = 10$. Therefore, the corresponding scene has excessive sparsity. In this case, the excessively high sparsity prior information constraint loses the description of the target details. With appropriate prior information parameter $\hat{\omega}$ estimated by the least squares estimator, the resolution-enhanced images of FLMC-SAR are obtained. In Fig. 10(d), all targets are successfully reconstructed, and the discontinuity of the image is also avoided. The surface target experiment shows that the proposed method can achieve the FLMC-SAR image resolution enhancement with a well description of the target details.

C. Real Data Experiment

In order to verify the effectiveness of the proposed method for FLMC-SAR imaging in practical applications, the real data experiment is carried out in this section.

The real data are collected by a K-band FLMC-SAR system, which works at spotlight mode with 8° beam azimuth width and 20° beam pitch width. The pulse repetition frequency (PRF) was 6000 Hz, and the synthetic aperture time was about 1.2 s containing 8192 pulses. The radar is equipped with a five-channel array antenna with a real aperture of 0.16 m, which is mounted onboard an airborne platform at the altitude of 4000 m and a speed of 80 m/s. A high-precision position and orientation system provides trajectory information during the data collection. The relevant system parameters of the system are shown in Table III.

The imaging result obtained by the SAR imaging is shown in Fig. 11(a). The vertical direction is north–south, and the horizontal direction is east–west. Due to the influence of motion error, the SAR image is defocused without motion compensation. Fortunately, this FLMC-SAR system is equipped with a high-precision position and orientation system, in which the accuracy of the gyroscope is less than 0.001°, and the accuracy of the accelerometer is less than $0.001g$ (where g is the gravitational acceleration). Therefore, fine motion compensation can be realized for the provided motion error to obtain the focused SAR image, as shown in Fig. 11(b). The Doppler ambiguity still exists, causing the imaging results of the left and right scenes aliased. According to the beamforming-based left–right Doppler ambiguity resolving method in [7] and [8], the FLMC-SAR image without ambiguity can be obtained, as shown in Fig. 11(c). Analyzing joint FLMC-SAR imaging and Doppler ambiguity, an FLMC-SAR image resolution enhancement method is proposed, which combines the signal sparsity with prior information regarding the spatial structure of the target. The imaging result via the proposed method is shown in Fig. 11(d). It is noteworthy that the array error is an important factor affecting the performance of the proposed method, and we have performed array deviation angle calibration according to the method in our previous work [7]. The projection of Fig. 11(c) and (d) in the geographic coordinate system is shown in Fig. 12(a) and (b), respectively. The satellite image from Google Earth is shown in Fig. 12(c). However, the limited azimuth resolution makes the structural details of the target in some areas loss, such as the area marked by the red rectangle in Fig 12(a). Compared to the satellite

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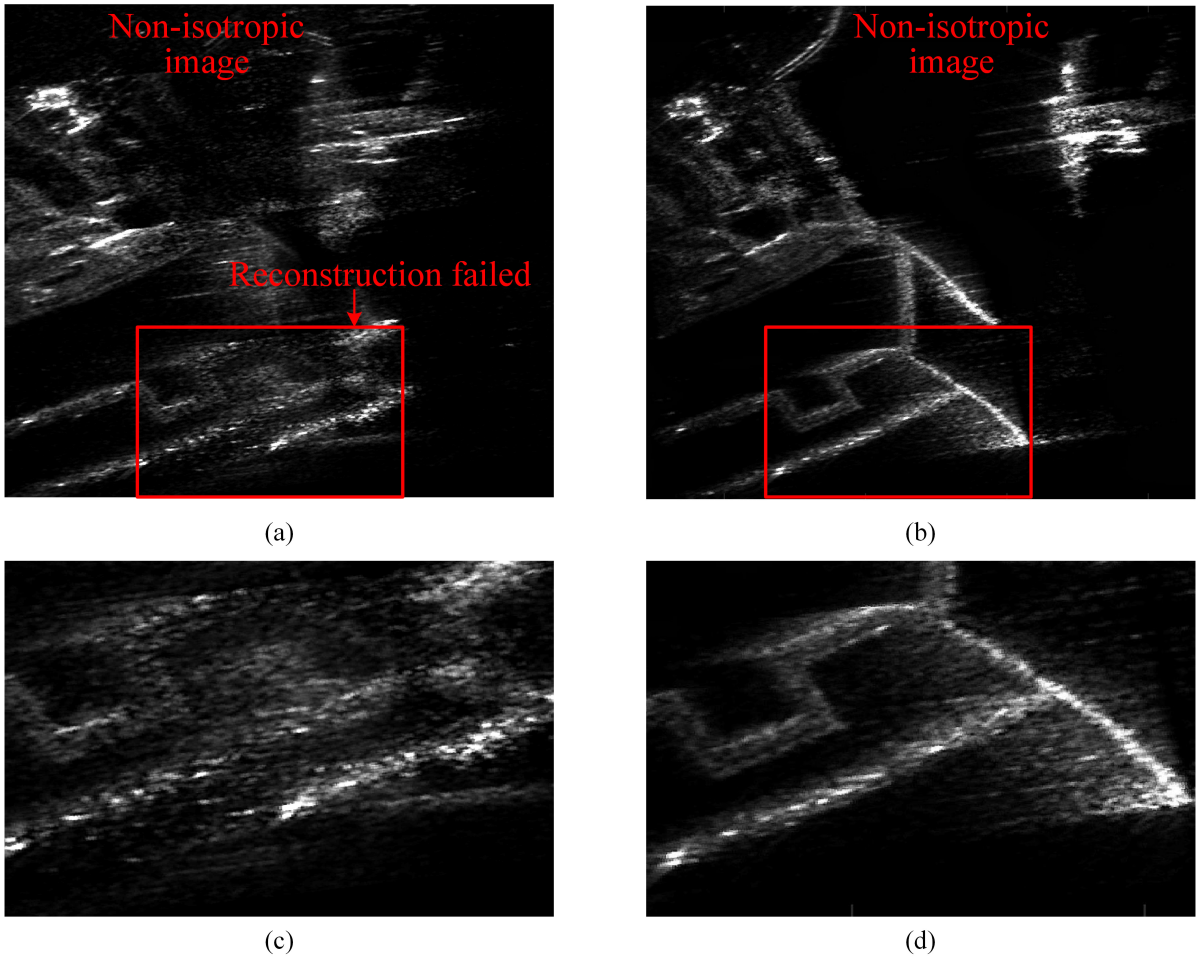

Fig. 13. Imaging results via different estimators. (a) Imaging results via traditional global least square estimator. (b) Imaging results via local least square estimator. (c) Magnified images of imaging results via traditional global least square estimator. (d) Magnified images of imaging results via local least square estimator.

image, the proposed method successfully obtained FLMC-SAR high-resolution imaging results without ambiguity and effectively enhanced the image resolution. The structural details of the target in the SAR image were successfully restored, and the magnified images marked by the red rectangle in Fig. 12(a) and (b) are shown in Fig. 12(d) and (e), respectively. The roads and coastline are clearly visible. To verify the effectiveness of the proposed algorithm, we give the east–west profile of the two roads in the selected north–south bin, as shown in Fig 12(f). Also, the amplitude of the azimuth pulse response function is normalized. The two roads can be well distinguished via the proposed method.

In the proposed method, the sparsity weight parameter is determined by the local least square estimator of the prior noise and signal statistics. We compare the imaging results of the traditional global least square estimator and that of the proposed local least square estimator, as shown in Fig. 13. Due to the nonisotropic image, the ℓ_1 constraint coefficient of global least square estimator cannot effectively reconstruct the whole image. In the area highlighted by the red rectangle in Fig. 13(a), the traditional global least squares cannot achieve accurate image reconstruction, resulting in the failure of resolution enhancement, as shown in Fig. 12(c). It means that the traditional global least square prior information parameters estimation is not suitable for FLMC-SAR nonisotropic imaging system. In the local least squares estimation, the ℓ_1 constraint coefficient is spatially related in the image domain. The spatial structure information of targets in the preliminary image can be transferred to the resolution enhancement and applied therein, as shown in Fig. 13(c). Different from conventional beamforming processing to resolve the FLMC-SAR left–right ambiguity, a Bayesian sparsity reconstruction optimization is developed to jointly ambiguity resolving and resolution enhancement in the azimuth angle image domain, which is significant for performance improvement.

V. Conclusion

The limited Doppler diversity of the targets in the forward-looking area and the inherent left–right Doppler ambiguity pose challenges for FLMC-SAR image processing. In this article, a sparsity-based FLMC-SAR image resolution enhancement method is proposed. First, the forward-looking area is preliminarily distinguished through the synthetic aperture. Therefore, the target signal in each resolution unit of

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the preliminary images is highly sparse. Then, the space–time observation model of FLMC-SAR is presented by analyzing joint FLMC-SAR imaging and Doppler ambiguity. The array's degrees of freedom are regarded as a key factor in signal sparse modeling and are also the key to solving the left–right Doppler ambiguity. A constrained optimization problem for FLMC-SAR image resolution enhancement is formulated in the FLMC-SAR image domain, which combines the signal sparsity with prior information regarding the spatial structure of the target. According to the preliminary imaging results of FLMC-SAR, a local least square estimator is developed to estimate the statistics of noise and target signal. Finally, the high-resolution FLMC-SAR images can be recovered from limited measurements. Experimental results based on both simulated and real data demonstrate that the proposed approach enhances the image resolution and recovers finer structure for FLMC-SAR image processing.

APPENDIX A
DERIVATION OF RCMC

Applying the range Fourier transform to (6), we get the target signal in the range frequency domain

$$\begin{aligned} &S_1(f_r, t_m, x, y, y_i) \\ &\quad = \mathrm{FT}[s_1(\tau, t_m, x, y, y_i)] \\ &\quad = A_P \cdot S(f_r) \cdot \exp\left[-j2\pi (f_c + f_r)\frac{R(t_m, x, y, y_i)}{c}\right] \end{aligned} \tag{39}$$

where $S(f_r) = \mathrm{FT}[\mathrm{sinc}(B\tau)]$, B is the bandwidth of LFM pulse, f_c is the carrier frequency, f_r is the range frequency, and c is the light speed.

Suppose that the point target with the beam pointing azimuth angle 0° and the reference slant range r_0 is selected as the reference target in the scene center. The coordinate of the reference target is $\mathbf{P}_0[x_0, y_0, 0] = [r_0 \sin\alpha, 0, 0]$. Then, the dechirped target signal can be expressed as

$$\begin{aligned} &S_1^1(f_r, t_m, x, y, y_i) \\ &= S_1(f_r, t_m, x, y, y_i) \cdot \exp\left[j2\pi (f_c + f_r)\frac{R(t_m, x_0, y_0, y_i)}{c}\right] \\ &= A_P \cdot S(f_r) \\ &\quad \cdot \exp\left\{-j2\pi (f_c + f_r) \cdot \frac{[R(t_m, x, y, y_i) - R(t_m, x_0, y_0, y_i)]}{c}\right\}. \end{aligned} \tag{40}$$

The Taylor expansion of $R(t_m, x, y, y_i) - R(t_m, x_0, y_0, y_i)$ is expressed as follows:

$$\begin{aligned} &R(t_m, x, y, y_i) - R(t_m, x_0, y_0, y_i) \\ &\quad \approx 2(r_P - r_0) - 2\sin\alpha v(\cos\theta - 1)t_m \\ &\qquad + \left(\frac{2v^2 - \sin^2\alpha v^2\cos^2\theta}{r_P} - \frac{2v^2 - \sin^2\alpha v^2}{r_0}\right)t_m^2. \end{aligned} \tag{41}$$

Then, (40) can be rewritten as

$$\begin{aligned} &S_1^1(f_r, t_m, x, y, y_i) \\ &\approx A_P \cdot S(f_r) \cdot \exp\left[-j2\pi (f_c + f_r)\frac{2(r_P - r_0)}{c}\right] \\ &\quad \cdot \exp\left\{-j2\pi (f_c + f_r)\frac{[-2\sin\alpha v(\cos\theta - 1)t_m]}{c}\right\} \\ &\quad \cdot \exp\left\{-j2\pi (f_c + f_r)\frac{\left[\left(\frac{2v^2 - \sin^2\alpha v^2\cos^2\theta}{r_P} - \frac{2v^2 - \sin^2\alpha v^2}{r_0}\right)t_m^2\right]}{c}\right\}. \end{aligned} \tag{42}$$

The first term in (40) is the range envelope of the signals, which determines the range unit that the target will focus on after range pulse compression. The second term in (40) is the first-order phase with respect to the radar slow time t_m. The third term in (40) is the second-order phase with respect to the radar slow time t_m.

After dechirping processing, the range curvature of the signal will not exceed one range cell, and thus, the linear RCM correction can be achieved through the first-order keystone transform. In order to scale the radar slow time, let

$$t_m = \frac{f_c}{f_c + f_r}\eta. \tag{43}$$

Substituting the above scale transform into (40), we can obtain

$$\begin{aligned} &S_1^1(f_r, \eta, x, y, y_i) \\ &\approx A_P \cdot S(f_r) \cdot \exp\left[-j2\pi (f_c + f_r)\frac{2(r_P - r_0)}{c}\right] \\ &\quad \cdot \exp\left\{-j2\pi f_c\frac{[-2\sin\alpha v(\cos\theta - 1)\eta]}{c}\right\} \\ &\quad \cdot \exp\left\{-j2\pi \frac{f_c^2}{f_c + f_r}\frac{\left[\left(\frac{2v^2 - \sin^2\alpha v^2\cos^2\theta}{r_P} - \frac{2v^2 - \sin^2\alpha v^2}{r_0}\right)\eta^2\right]}{c}\right\}. \end{aligned} \tag{44}$$

Applying the range Fourier transform to (44), we can obtain the signal after RCMC

$$\begin{aligned} &s_1(\tau, t_m, x, y, y_i) \\ &\quad = A_P \cdot \mathrm{sinc}\left[B\left(\tau - \frac{2(r_P - r_0)}{c}\right)\right] \\ &\qquad \cdot \exp\left[-j\frac{2\pi}{\lambda}[R(t_m, x, y, y_i) - R(t_m, x_0, y_0, y_i)]\right]. \end{aligned} \tag{45}$$

APPENDIX B
DERIVATION OF CONJUGATE GRADIENT

In order to avoid the nondifferentiability of the ℓ_1 norm in (37), we use the following smooth approximation [42]:

$$\|\hat{\mathbf{A}}(x, y)\|_1 = \sum_{i=1}^{N} (a^2(x, yx, y) + \xi)^{\frac{1}{2}} \tag{46}$$

where ξ is a small nonnegative constant and $a(x, y)$ is the element in $\mathbf{A}(x, y)$. The Bayesian kernel is slightly modified as

$$\begin{aligned} \hat{\mathbf{A}}(x, y) = \arg\min \Bigg[&\|\mathbf{S}_4(x, y) - \mathbf{H}\hat{\mathbf{A}}(x, y)\|_F^2 \\ &+ \hat{\omega}(x, y)\sum_{i=1}^{N}\left(a^2(x, y) + \xi\right)^{\frac{1}{2}}\Bigg]. \end{aligned} \tag{47}$$

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Then, the conjugate gradient of cost function in (47) over $\mathbf{A}(x, y)$ can be obtained as

$$\nabla f(\hat{\mathbf{A}}(x, y)) = \mathbf{H}(\mathbf{A}(x, y)) \cdot \mathbf{A}(x, y) - 2\mathbf{H}^H \mathbf{S}_4(x, y) \quad (48)$$

where the Hessian matrix can be approximated as follows:

$$\mathbf{H}(\mathbf{A}(x, y)) = 2\mathbf{H}^H \mathbf{H} + \hat{\omega}(x, y) \cdot \mathbf{U}(\mathbf{A}(x, y)). \quad (49)$$

In this way, we can obtain the iteration of the quasi-Newton solver.

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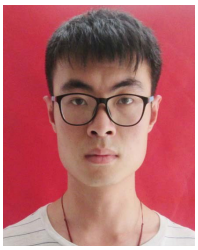

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